

R E V I E W

Efficacy and safety of novel beta-lactam/beta-lactamase inhibitor combinations in multidrug-resistant gram-negative bacterial infections: A systematic review and meta-analysis

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ABSTRACT

Background and aim: Multidrug-resistant Gram-negative bacterial infections (MDR-GNB) pose a significant global health challenge, limiting the effectivity of standard antibiotics treatment and contributing to increased morbidity and mortality. Beta-lactam/beta-lactamase inhibitor (BL/BLI) combinations have emerged as promising alternatives to standard antibiotic therapy. This systematic review and meta-analysis aim to evaluate the efficacy and safety of BL/BLI combinations in MDR-GNB infections.

Methods: A comprehensive literature search was conducted, including clinical trials comparing BL/BLI combinations with standard antibiotic therapy or placebo. Primary outcomes included clinical cure rates (CCR), microbiological eradication, mortality, adverse events, and survival rates. The risk of bias was assessed using the Revised Tool for Risk of Bias in Randomized Trials (RoB 2.0). Meta-analysis was performed using Review Manager 5.4, and heterogeneity was quantified using the I² statistic.



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Results: Seventeen studies assessed CCR, demonstrating a significant advantage for BL/BLI combinations (Odds ratio (OR) 2.43 (95% CI 1.95–3.03)). Thirteen studies reported microbiological eradication, favoring the intervention group (OR 2.45(1.42–4.24)). Twenty studies analyzed mortality, revealing a lower risk in the intervention group (OR 0.44(0.31–0.64)). Eight studies examined adverse events, indicating a lower incidence in the BL/BLI group (OR 0.60(0.40–0.91)). Three studies evaluated survival rates, showing a significant benefit with BL/BLI combinations (OR 2.02(1.10–3.71)).

Conclusions: BL/BLI combinations in MDR-GNB infections significantly improved clinical cure rates, microbiological eradication, and survival. It also reduced mortality and adverse events, highlighting its superiority over standard therapy in MDR-GNB. (www.actabiomedica.it)

Key words: multidrug-resistant gram-negative bacteria, beta-lactamase inhibitors, microbiological eradication, infectious disease

Introduction

Antimicrobial resistance (AMR) is a critical global health threat, with bacterial resistance contributing to rising morbidity, mortality, and healthcare costs. The World Health Organization (WHO) reported that bacterial AMR directly caused 1.27 million deaths worldwide in 2019, with projections estimating up to 10 million deaths annually by 2050 if no effective measures are taken to combat its spread (1). The overuse and misuse of antimicrobials in both human and veterinary medicine have accelerated the emergence and dissemination of multidrug-resistant pathogens, leading to a growing crisis in infection management (1). Beyond its clinical impact, AMR poses significant economic burdens, with potential global GDP losses ranging from US\$ 1 trillion to US\$ 3.4 trillion by 2030. Among the most concerning AMR threats are multidrug-resistant Gram-negative bacteria (MDR-GNB), particularly carbapenem-resistant Enterobacterales (e.g., *Klebsiella pneumoniae*, *Escherichia coli*), multidrug-resistant *Pseudomonas aeruginosa*, and carbapenem-resistant *Acinetobacter baumannii* (2). These MDR-GNB pathogens have, employ various mechanisms such as the production of β -lactamases, efflux pumps, and alterations in membrane permeability, to evade treatment, rendering conventional and newer generation antibiotics like extended-spectrum cephalosporins and carbapenems ineffective (3,4).

Given their rapid resistance acquisition and global spread, MDR-GNB infections have been designated as a top-priority pathogen by the WHO, emphasizing the urgent need for novel therapeutic strategies. Beta-lactam antibiotics are among the most widely used antimicrobials and remain the cornerstone of antimicrobial therapy in clinical practice. These agents function by inhibiting the transpeptidase activity of penicillin-binding proteins (PBPs), which are crucial for peptidoglycan (PG) biosynthesis, the primary structural component of bacterial cell walls (5,6). The inhibition of PBPs by β -lactams disrupts the delicate balance between PG hydrolases and PG synthases, ultimately compromising cell wall integrity and leading to bacterial cell lysis and death through the activation of autolysins (7,8). However, the widespread use of β -lactams has been accompanied by the emergence of β -lactamase enzymes, which hydrolyze and inactivate these antibiotics, rendering them ineffective against resistant bacterial strains. To overcome this challenge, β -lactamase inhibitors (BLIs) have been developed to restore the activity of β -lactam antibiotics by inhibiting β -lactamase enzymes. By combining β -lactams with BLIs, it is possible to enhance their antimicrobial efficacy and expand their spectrum of activity against MDR-GNB. In response to the growing AMR crisis, several novel β -lactam/ β -lactamase inhibitor (BL/BLI) combinations have been introduced to combat MDR-GNB infections. Among the most promising agents

are ceftazidime/avibactam, ceftolozane/tazobactam, and meropenem/vaborbactam, which have demonstrated potent activity against resistant Gram-negative pathogens (9,10). These novel combinations have been evaluated in large-scale randomized controlled trials (RCTs) and have shown efficacy in treating complicated infections, including complicated urinary tract infections (cUTI), complicated intra-abdominal infections (cIAI), and hospital-acquired pneumonia/ventilator-associated pneumonia (HAP/VAP). Despite their clinical potential, evidence on the overall efficacy and safety of these new BL/BLI combinations remains limited, particularly for MDR-GNB infections. Given the increasing prevalence of MDR-GNB and the urgent need for effective treatment options, this systematic review and meta-analysis aim to comprehensively evaluate the efficacy and safety of novel BL/BLI combinations in managing MDR-GNB infections. By analyzing available clinical data, this study seeks to provide a clearer understanding of the role of these agents in improving patient outcomes, addressing gaps in the current literature, and informing future antimicrobial stewardship efforts.

Methods

Protocol registration

This meta-analysis was conducted based on the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement guidelines (11). This study was registered in PROSPERO with registration number CRD42025643343.

Search strategy

The literature search was carried out on six databases, including Cochrane Library, Science Direct, PubMed, Scopus, ProQuest, and BMJ Journals, until January 2025. The literature search was done with keywords using Boolean operators as detailed in Table S1.

Study eligibility criteria

The inclusion and exclusion criteria were pre-defined before the literature search to ensure the

homogeneity of included studies. The inclusion criteria were 1) data available or accessible in the English language, 2) clinical trial studies, 3) studies involving patients multidrug-resistant Gram-negative bacterial infections, and 4) studies must compare outcomes between individuals with novel beta-lactam or beta-lactamase inhibitor combinations and standard antibiotic therapy or placebo, reporting on odds ratios for the association between clinical cure rates, microbiological eradication, mortality, and resistance emergence. Conversely, studies were excluded based on 1) non-human sampling and 2) non-peer-reviewed studies. The authors individually assessed the eligibility of each study, and any disagreements were resolved through discussion.

Outcome measures

The impact of novel beta-lactam/beta-lactamase inhibitor combinations in multidrug-resistant Gram-negative bacterial infections is evaluated in this study using three efficacy parameters and two safety parameters. The efficacy parameters include the Clinical Cure Rates (CCR) related to Beta-Lactam/Beta-Lactamase Inhibitor (BL/BLI) combinations and the extent of microbiological eradication produced. The safety parameter focuses on Adverse events, with an emphasis on patient survival rates. These indicators are intended to give a thorough grasp of the function of beta-lactamase inhibitors in MDR-GNB. Results from the included papers were independently extracted by each author for the quantitative analysis, and discrepancies were discussed and resolved.

Quality assessment

The Revised Tool for Risk of Bias in Randomized Trials (RoB 2.0) was used to evaluate the bias risk in the included studies (12). Subsequently, the assessment findings were recorded in the “bias” section of a Microsoft Excel 2021 spreadsheet. The spreadsheet was then posted to the ROBVIS website (<https://mcguinlu.shinyapps.io/robvis/> (Accessed on February 2025)), where a traffic light system was used to visually represent the assessment’s findings (13). This systematic approach ensured a comprehensive and transparent depiction of bias levels in the included studies.

Statistical analysis

The evaluation of bias risk in the included studies utilized the Revised Tool for Risk of Bias in Randomized Trials (RoB 2.0) (12). The assessment results were recorded in a Microsoft Excel 2021 spreadsheet's "bias" section. Following this, the spreadsheet was uploaded to the ROBVIS website (<https://mcguinlu.shinyapps.io/robvis/>; accessed in February 2025), where the assessment results were presented visually using a traffic-light system. The levels of bias in the included studies were fully and openly represented as a result of this systematic approach. The Meta-analysis was conducted by Review Manager 5.4.

Results

Study selection and characteristics

The elimination of 174 duplicate studies prior to screening yielded 3,291 articles for comprehensive inclusion assessment. A total of 3,175 articles were excluded: 1,720 with irrelevant titles and abstracts and 1,455 reviews, letters to the editor, conference papers, or books. Subsequently, 89 full-text articles were sought for retrieval. However, 32 articles could not be retrieved. Among the 57 reports assessed for eligibility, 35 studies were excluded because their outcomes were irrelevant. As shown in Figure 1, the final assessment included 22 studies in this systematic review. The demography and clinical characteristics of each study were examined and listed in Table 1.

Quality appraisal

Seventeen studies were at low risk of confounding bias, while four were at moderate risk (Figure 2). Twenty studies were at low risk, and one was missing information due to participant selection of participants. Nineteen studies were at low risk of bias due to the classification of interventions; one was at moderate risk; and one study was missing information. Two studies had missing information due to bias caused by deviations from intended interventions, while nineteen studies were of low risk. All studies except one were of low risk for bias due to missing

data. All studies were of low risk of bias in the measurement of outcome. One study was of serious risk, one study of moderate risk and the other nineteen studies were of low risk under bias in the selection of reported results. The overall result of the assessment shows about 75% low risk of bias, about 5% on serious risk, and about 20% on moderate risk.

Clinical Cure Rates (CCR)

Seventeen studies have reported clinical cure rates (CCR) and results, as shown in Figure 3A, indicating that the intervention group is favorable, with an OR of 2.43 (95% CI 1.95-3.03, $p < 0.0001$). P-value revealed that the result of the study is statistically significant. Heterogeneity was very low ($I^2 = 17.1\%$), and the funnel plot showed no outliers, indicating true homogeneity among the studies (Figure 3B).

Microbiological eradication

Thirteen studies have reported microbiological eradication. The results, as seen in Figure 4A, showed that the intervention group is favorable with OR of 2.45 (95% CI 1.42; 4.24, $p=0.0013$). The p-value revealed that the result is statistically significant. Heterogeneity ($I^2 = 79.5\%$) and the funnel plot also show outliers indicating evidence of true heterogeneity among the studies (Figure 4B).

Mortality

Twenty studies reported mortality and the results as seen in Figure 5A showed that the control group is favorable with OR of 0.44 (95% CI 0.31; 0.64, $p<0.0001$). The p-value revealed that the result is statistically significant. Heterogeneity was moderate ($I^2=65.3\%$), and the funnel plot also shows few outliers, indicating evidence of true heterogeneity among the studies (Figure 5B).

Adverse events

Eight studies reported Adverse events. The results, as seen in Figure 6A, showed that the control group is favorable with an OR of 0.60 (95% CI 0.40;

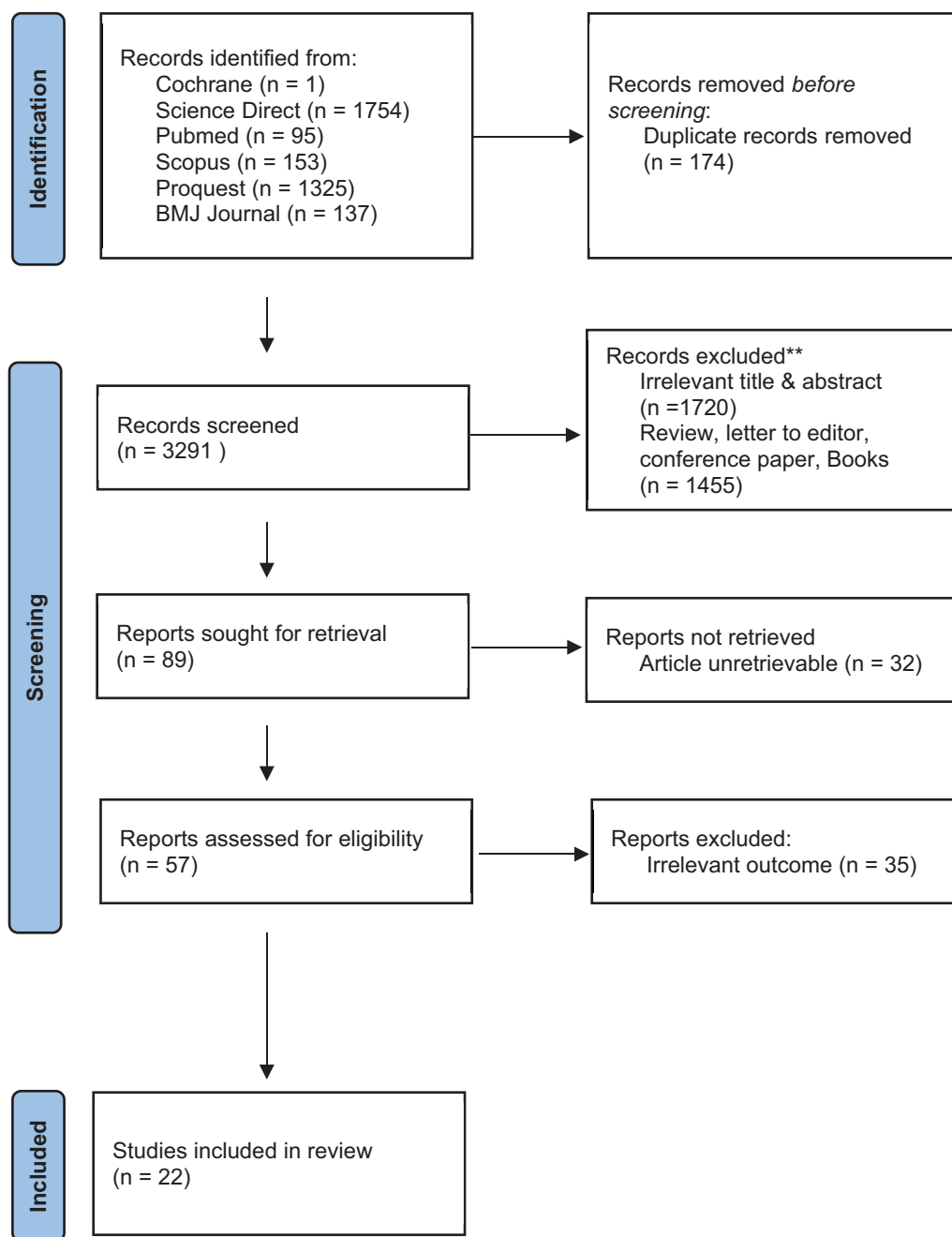


Figure 1. Preferred reporting items for systematic reviews and meta-analyses (PRISMA) flowchart for study identification and selection.

0.91, $p=0.0161$). The p -value revealed that the result is statistically significant. Heterogeneity was low ($I^2=9.7\%$). The funnel plot also shows no outliers indicating evidence of true homogeneity among the studies (Figure 6B).

Survival rates

Three studies reported survival rates and the results as seen in Figure 7A showed that the intervention group is favorable with OR of 2.02 (95% CI 1.10;

Table 1. Baseline characteristics of included studies

Author	Year	Country	Type of Study	Total Sample (n)	Male (n)	Female (n)	Intervention (n)	Control (n)	Intervention	Control	Age group Mean (SD)
Mogyorodi et al.	2022	Hungary	Retrospective, observational	51	34	17	18	33	Ceftolozane/tazobactam (C/T)	colistimethate sodium (CMS)	66 (6)
Chen et al.	2021	China	Retrospective, observational	187	115	72	73	114	CAZ-AVI	non CAZ-AVI	67 (14.15)
Karaiskos et al.	2020	Greece	Prospective observational	142	95	47	71	71	ceftazidime/avibactam-based regimens	Other regimens	60.9 (17.1)
Gu et al.	2021	China	Retrospective observational	90	63	27	42	48	ceftazidime-avibactam	other active antibiotics	61.05 (16.65)
Shields et al.	2017	USA	Observational, retrospective	109	61	48	68	41	ceftazidime-avibactam	other active antibiotics	61 (25 to 91)
Tsolaki et al.	2020	Greece	Observational, retrospective	77	56	21	41	36	ceftazidime-avibactam	control	60.51 (2.55)
Shi et al.	2021	China	Retrospective	105	82	23	43	62	ceftazidime-avibactam (CAZ-AVI)	tigecycline (TGC)-based regimen	61.65 (18.2)
Castón et al.	2017	Spain and Israel	Retrospective, observational	31	19	12	8	23	ceftazidime-avibactam	other active antibiotics	59 (6)
Duin et al.	2017	USA	Prospective, observational	137	65	72	38	99	ceftazidime-avibactam	Colistin	61 (12)
Hakeam et al.	2021	Saudi Arabia	Retrospective, cohort	61	36	25	32	29	ceftazidime-avibactam	Colistin	53.7 (19.3)
Fang et al.	2021	China	Retrospective, observational	115	88	27	37	78	ceftazidime-avibactam	Polymyxin B	63 (11.5)
Zhang et al.	2021	China	Retrospective	54	32	22	22	32	ceftazidime-avibactam	control	39.15 (9.9)
Falcone et al.	2020	Italy	Prospective, observational	102	71	31	52	50	ceftazidime-avibactam + aztreonam	other active antibiotic	70 (55-78)

Tumbarello et al.	2018	Italy	Retrospective, observational	208	135	73	104	104	104	ceftazidime-avibactam	other active antibiotic	66 (40-82)
Yu et al.	2024	Taiwan	Retrospective, cohort	142	83	59	82	82	60	ceftazidime-avibactam	other active antibiotic	73.5 (64.25-80.87)
Almangour et al.	2022	Saudi Arabia	Retrospective, cohort	230	143	87	149	149	81	ceftazidime-avibactam	Colistin	58 (18)
Chen et al.	2022	China	Retrospective, observational	136	100	36	51	51	85	ceftazidime-avibactam	polymyxin B	61.5 (51.25-70)
Liu et al.	2024	China	Retrospective cohort	298	272	26	116	116	182	ceftazidime-avibactam	polymyxin B	60.5 (18)
Gurlevik et al.	2025	Turkey	Retrospective, observational	136	74	62	76	76	60	meropenem	Colistin	NR
Hakeam et al.	2022	Saudi Arabia	Retrospective, cohort	46	23	23	17	17	29	Ceftolozane-tazobactam	Colistin	53.2 (20.1)
Harrera et al.	2024	Argentina	Prospective, observation	70	47	23	18	18	52	Ceftazidime-avibactam	other active antibiotic	NR

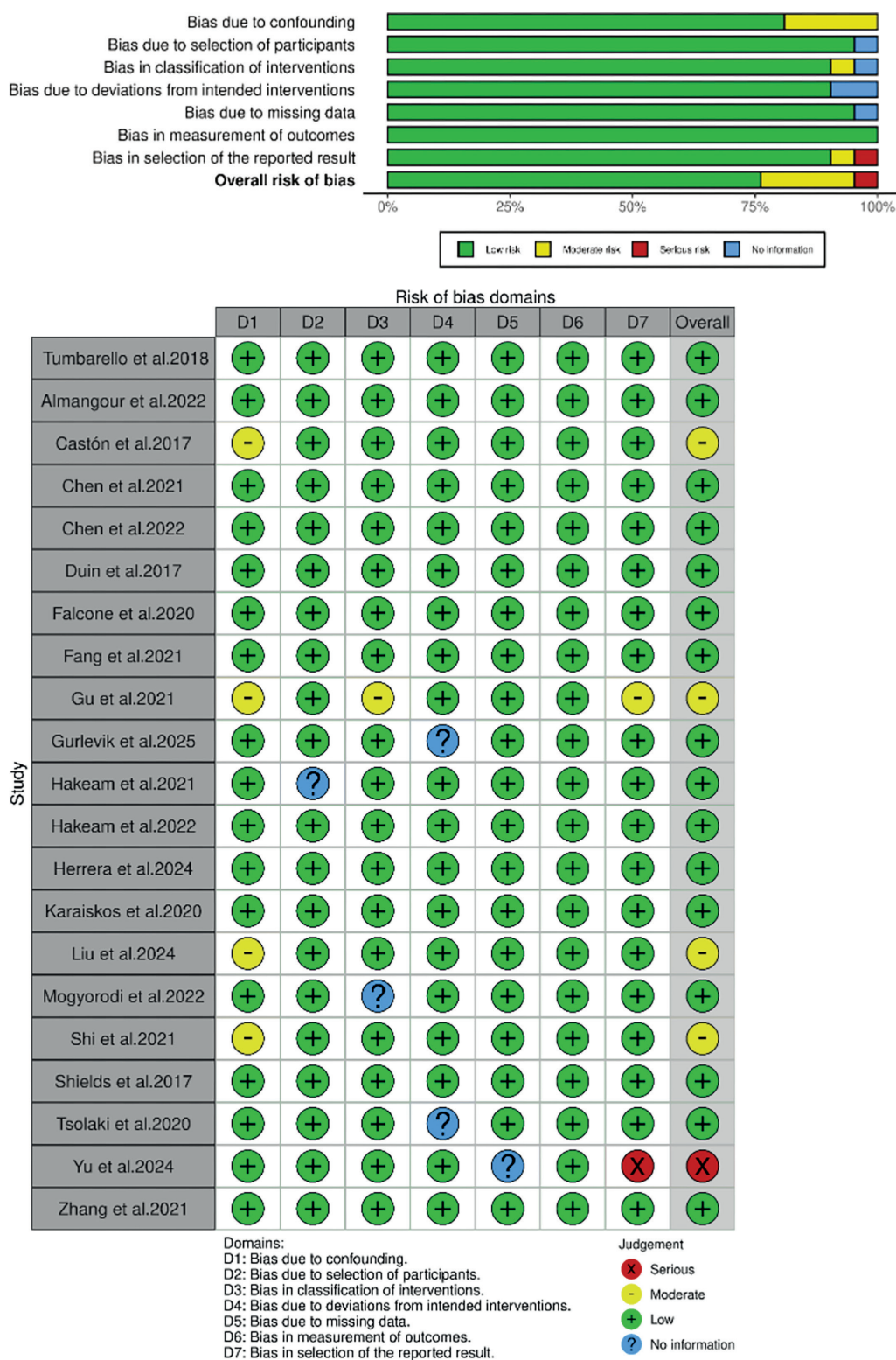


Figure 2. The assessment of bias risk in the included studies was conducted using the Revised Tool for Risk of Bias in Non-Randomized Studies-of interventions (RoBINS-i) for the twenty-one non-randomized studies included in this study.

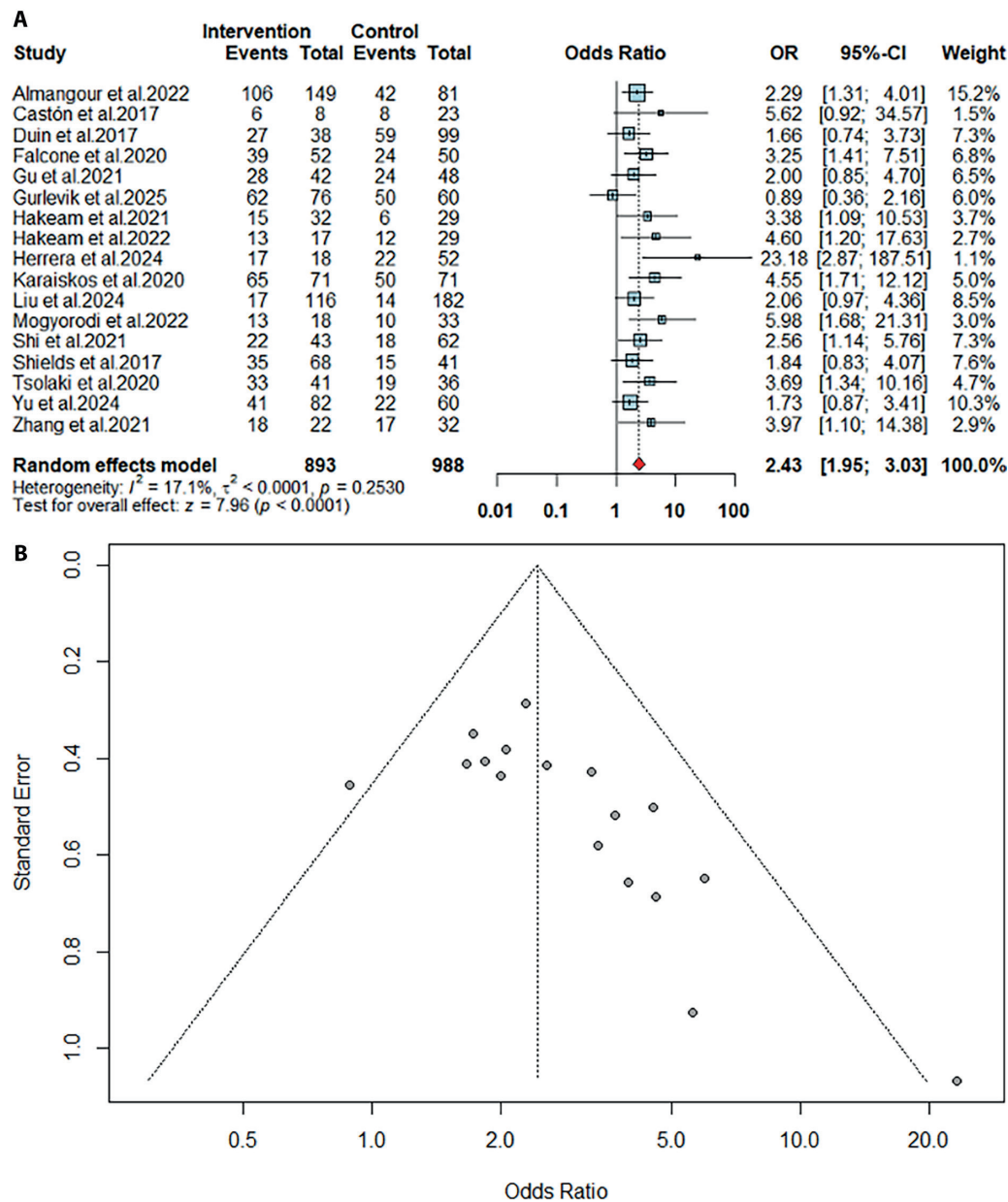


Figure 3. Analysis of clinical cure rates (CCR). (A) Forest plot that compares the odds ratio of CCR in MDR-GNB treated with novel BL/BLI combinations vs. the control. The blue square and solid lines represent the odds ratio with 95% confidence intervals. The size of the squares indicates the weight of each study. The black rhombus indicates the pooled estimate with 95% confidence intervals. (B) Funnel plot of included studies in CCR analysis. BL/BLI, Beta-lactam/beta-lactamase inhibitor; MDR-GNB, Multidrug-resistant Gram-negative bacterial infections.

3.71, $p=0.0242$) and the p -value revealed that the result is statistically significant. Heterogeneity was low ($I^2=29.5\%$), and the funnel plot also shows no outliers, indicating evidence of true homogeneity among the studies (Figure 7B).

Discussion

This systematic review and meta-analysis provide robust evidence that novel BL/BLI combinations significantly improve clinical outcomes in MDR-GNB

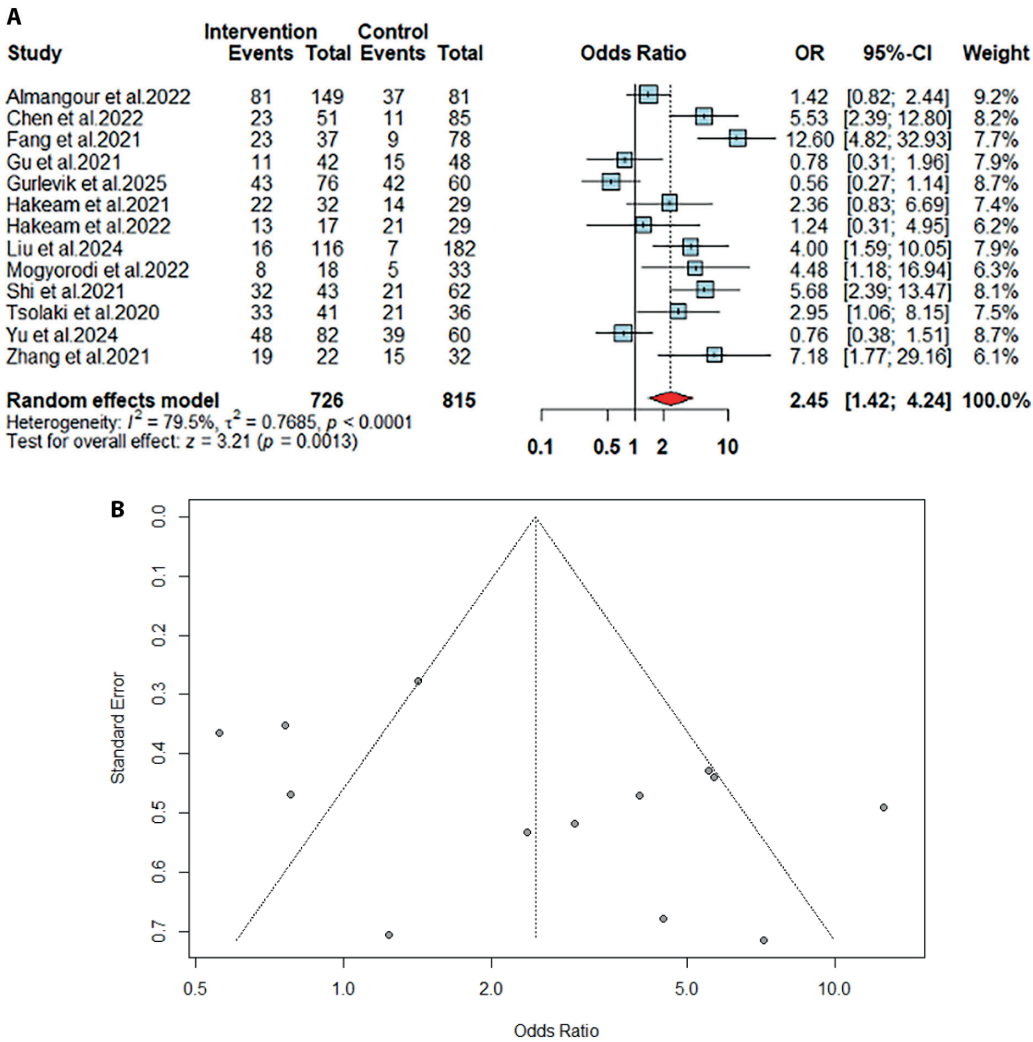


Figure 4. Analysis of microbiological eradication. (A) Forest plot that compares the odds ratio of microbiological eradication in MDR-GNB treated with novel BL/BLI combinations vs. the control. (B) Funnel plot of included studies in microbiological eradication analysis. BL/BLI, Beta-lactam/beta-lactamase inhibitor; MDR-GNB, Multidrug-resistant Gram-negative bacterial infections.

infections. In this meta-analysis of 22 observational studies, we found that the use of BL/BLI combinations was associated with improved clinical cure rates, greater microbiological eradication, lower mortality, increased survival rates and fewer adverse event rates among patients with MDR-GNB infections. These results highlight the clinical relevance of BL/BLI combinations in combating MDR-GNB infections, an urgent global health concern. MDR-GNB infections pose a major global health challenge, as resistance to commonly used antibiotics continues to rise. The increasing prevalence

of β -lactamases, including carbapenemases, extended-spectrum β -lactamases (ESBL), AmpC β -lactamases, and metallo- β -lactamases, has significantly reduced the efficacy of conventional antimicrobial agents. This has relegated certain antibiotic classes, such as polymyxins, to last-resort options despite their associated toxicity and unfavorable pharmacokinetic properties. The development of BLI combinations has emerged as a promising strategy due to their broad-spectrum β -lactamase inhibition, high enzymatic affinity, minimal toxicity, and favorable pharmacokinetics (14,15).

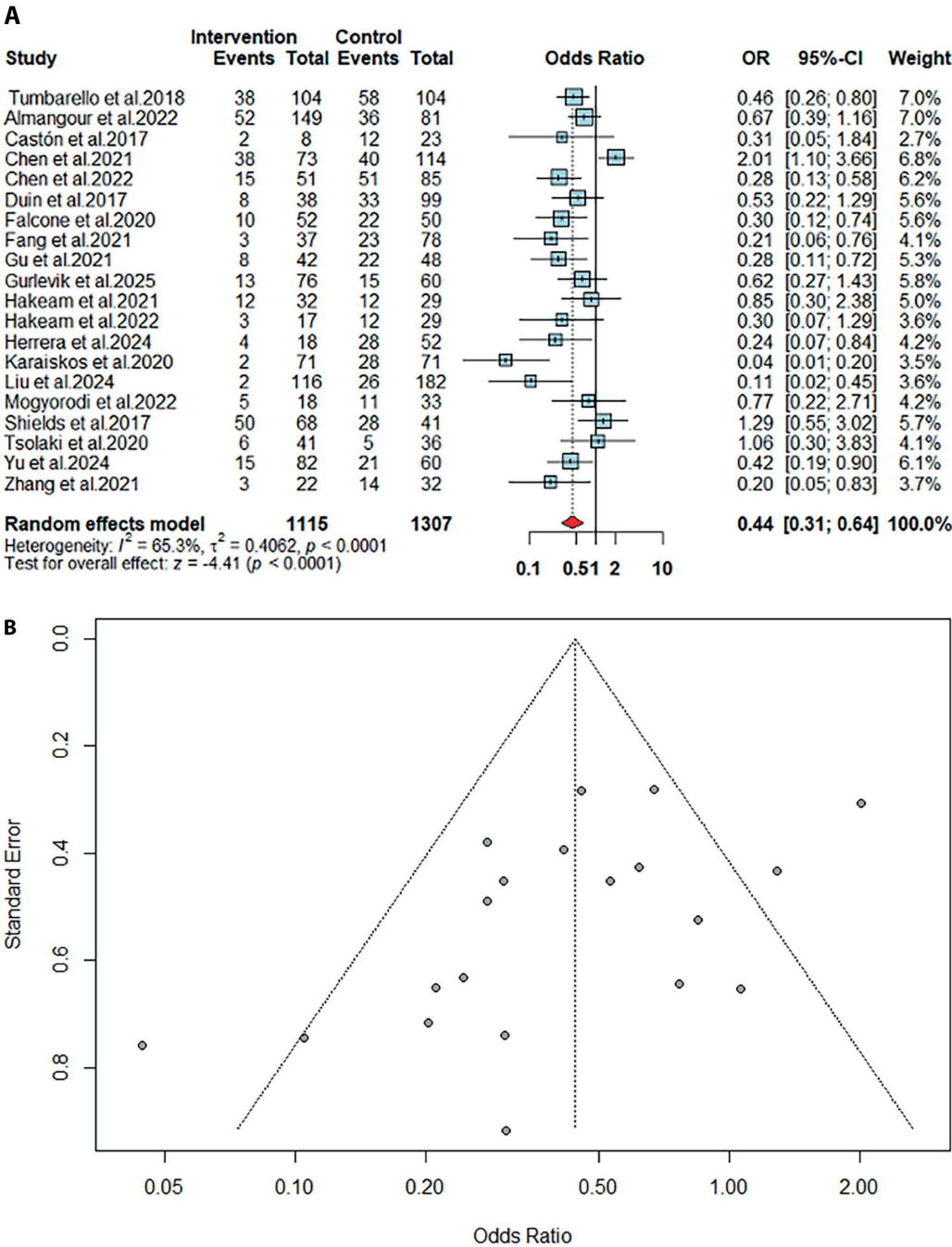


Figure 5. Analysis of mortality. (A) Forest plot that compares the odds ratio of mortality in MDR-GNB treated with novel BL/BLI combinations vs. the control. (B) Funnel plot of included studies in mortality analysis. BL/BLI, Beta-lactam/beta-lactamase inhibitor; MDR-GNB, Multidrug-resistant Gram-negative bacterial infections.

The significant increase in clinical cure rates (OR = 2.43 (95% CI 1.95; 3.03), $p < 0.0001$) underscores the efficacy of BL/BLI combinations in resolving infections. This aligns with prior studies demonstrating that

β -lactamase inhibitors restore β -lactam antibiotic activity against resistant pathogens by neutralizing enzymatic degradation (Reference). The low heterogeneity ($I^2 = 17.1\%$) supports the reliability and consistency of

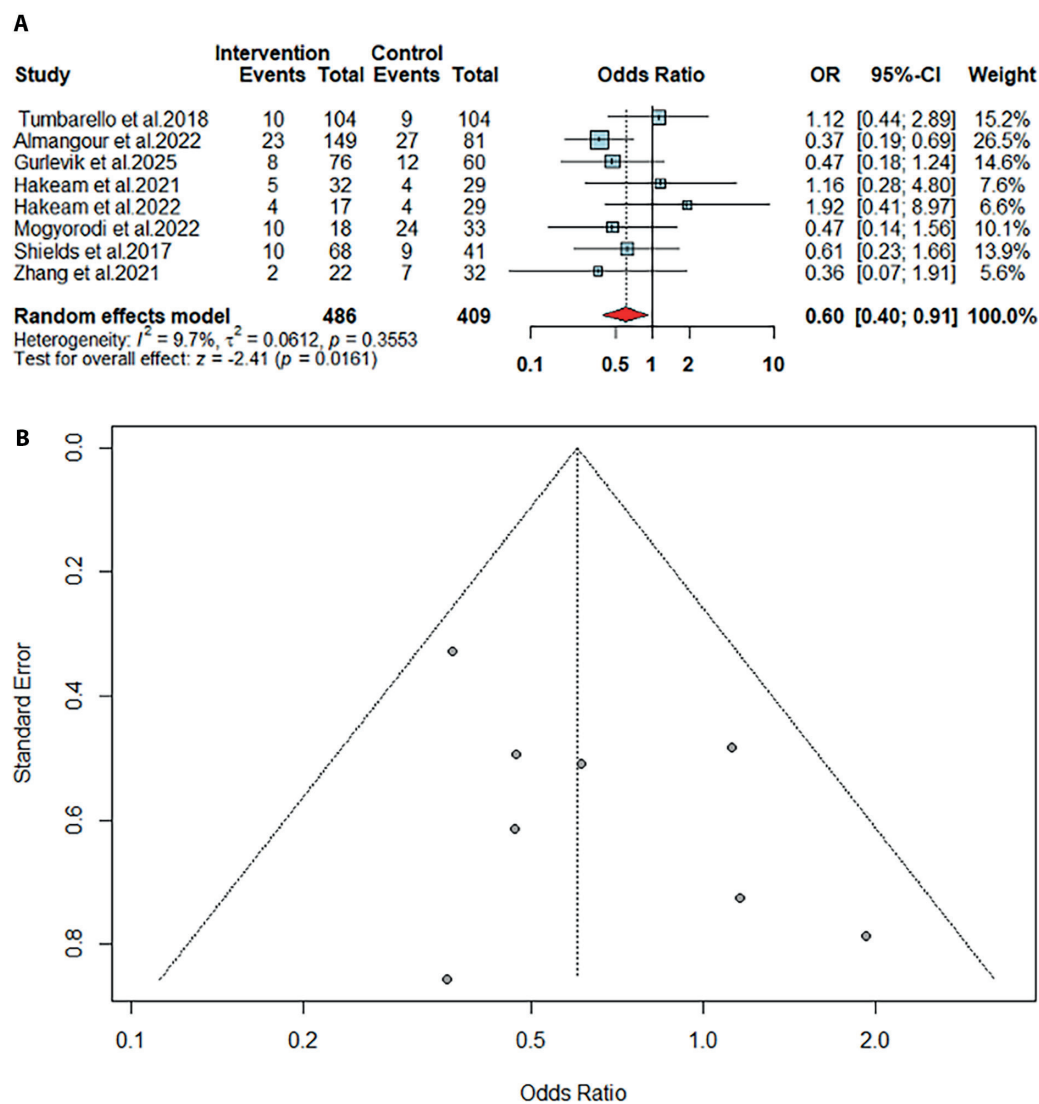


Figure 6. Analysis of adverse events. (A) Forest plot that compares the odds ratio of adverse events in MDR-GNB treated with novel BL/BLI combinations vs. the control. (B) Funnel plot of included studies in adverse events analysis. BL/BLI, Beta-lactam/beta-lactamase inhibitor; MDR-GNB, Multidrug-resistant Gram-negative bacterial infections.

this finding. Among the 17 studies analyzed, most favored the intervention group, although one study (16) did not find a significant benefit (OR = 0.89 (95% CI 0.36; 2.16)). These results are supported by previous meta-analyses comparing ceftazidime-avibactam (CAZ-AVI) with standard therapy, which found superior CCR in the BLI group (17). The microbiological eradication rate was also significantly higher in the BLI group than in the control group ($p = 0.0013$), consistent with findings from previous meta-analyses (18).

Avibactam, for example, restores ceftazidime’s activity by protecting it from enzymatic degradation, thereby enhancing bacterial eradication. This mechanism ensures prolonged bactericidal activity, even against carbapenem-resistant and other highly resistant strains (19). The observed reduction in mortality (OR = 0.44, 95% CI 0.31–0.64, $p < 0.0001$) suggests that BL/BLI combinations provide a survival advantage, potentially due to their enhanced bactericidal activity. This finding is consistent with prior meta-analyses reporting

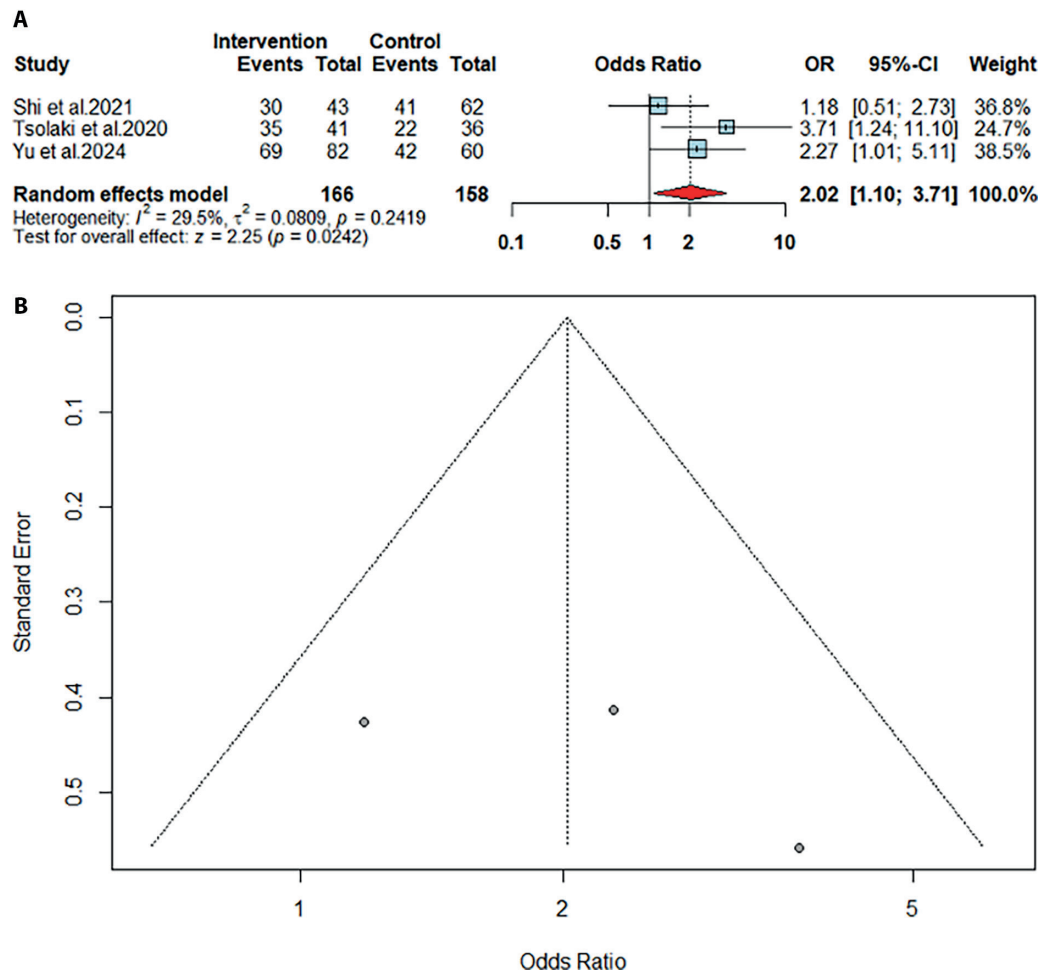


Figure 7. Analysis of survival. (A) Forest plot that compares the odds ratio of survival in MDR-GNB treated with novel BL/BLI combinations vs. the control. (B) Funnel plot of included studies in survival analysis. BL/BLI, Beta-lactam/beta-lactamase inhibitor; MDR-GNB, Multidrug-resistant Gram-negative bacterial infections.

reduced mortality with BLI-based regimens (17,20). This result is particularly relevant given the global rise in carbapenem-resistant *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*, and *Acinetobacter baumannii*, in which limited treatment options often lead to poor clinical outcomes. Furthermore, our analysis indicated that BLI combinations were associated with a higher survival rate compared to the control group. Previous studies have similarly demonstrated superior 28-day survival rates in patients receiving CAZ-AVI (85.4%) versus those in the control group (61.1%) (21). Another study reported that the use of CAZ-AVI increased survival by nearly 30% in a cohort of critically

ill patients (22). The improved survival observed in our analysis is likely driven by the combination of high clinical cure rates and reduced mortality. The low heterogeneity ($I^2 = 29.5\%$) indicates robust and reliable findings, although the relatively small number of studies necessitates further research to confirm these benefits. A key advantage of BL/BLI combinations is their favorable safety profile. Our analysis found a lower incidence of adverse events in the intervention group. This contrasts with conventional last-resort antibiotics such as polymyxins and aminoglycosides, which are associated with nephrotoxicity and neurotoxicity. However, previous meta-analyses found no

significant difference in adverse events between BLI combinations and standard therapy (23,24). The most commonly reported adverse effect associated with BLI use was diarrhea, followed by anemia and hypokalemia (25,26). Despite this discrepancy with prior studies, our analysis demonstrated low heterogeneity ($I^2 = 9.7\%$), suggesting that the findings are robust and may be generalizable to a wider population.

Despite these promising findings, heterogeneity was moderate to high in some outcomes, particularly microbiological eradication ($I^2 = 79.5\%$) and mortality ($I^2 = 65.3\%$), indicating variations in study populations, resistance mechanisms, and treatment protocols. While our subgroup analyses attempted to account for these differences, further research is needed to optimize BL/BLI use across diverse patient settings. The presence of outlier studies also limits the generalizability of these results to broader clinical settings. Several limitations should be acknowledged. First, although this study included RCTs and well-designed observational studies, the predominance of non-randomized research increases susceptibility to confounding bias. Second, significant heterogeneity was observed in microbiological eradication and mortality outcomes, limiting the generalizability of these findings. Third, the number of studies evaluating survival rates was relatively small, preventing robust subgroup analyses. Additionally, variations in study design, antibiotic comparators, and resistance mechanisms may have influenced the pooled estimates. Lastly, cost-effectiveness analyses were not within the scope of this review but remain a crucial consideration for real-world implementation.

Conclusion

Our meta-analysis suggests that novel BL/BLI combinations significantly improve clinical outcomes in MDR-GNB infections by enhancing cure rates, promoting bacterial eradication, reducing mortality, and increasing survival. Additionally, these agents also demonstrate a favorable safety profile, making them a valuable addition to the antimicrobial arsenal. While the findings support the clinical use of BLI combinations, the observed heterogeneity in some outcomes

highlights the need for further research, particularly on pathogen-specific efficacy, long-term clinical impact, and cost-effectiveness. As antimicrobial resistance continues to escalate, integrating BL/BLI combinations into treatment protocols should be prioritized to improve patient outcomes and support antimicrobial stewardship initiatives.

Conflict of Interest: Each author declares that he or she has no commercial associations (e.g. consultancies, stock ownership, equity interest, patent/licensing arrangement etc.) that might pose a conflict of interest in connection with the submitted article.

Authors Contribution: RNR, SZ, DDCHR, ADJR, MNY, RJ, AFR, KWAS, AE, MVA contributed to the literature search, study screening, data extraction, data curation, and drafting of the original manuscript. MAE, HdS, INK contributed to the conceptualization, methodology, data acquisition, data analysis, data interpretation, drafting of the original manuscript, and critical revision of the manuscript for important intellectual content. INK contributed to study supervision, study design, and validation. All authors made substantial contributions to the work, participated in data interpretation and manuscript preparation or revision, approved the final version of the manuscript for submission/publication, and agree to be accountable for all aspects of the work.

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Annex

Supplementary file

Table S1. Literature search terms

Cochrane	("multidrug resistant" OR "MDR" OR "drug resistant bacteria" OR "antibiotic resistant") AND ("infection" OR "bloodstream infection" OR "sepsis" OR "pneumonia" OR "ventilator-associated pneumonia" OR "urinary tract infection" OR "catheter infection" OR "catheter-related infection") AND ("beta-lactamase inhibitor" OR "beta-lactam/beta-lactamase inhibitor" OR "ceftazidime-avibactam" OR "avibactam" OR "ceftazidime" OR "aztreonam-avibactam" OR "tazobactam" OR "imipenem-cilastatin" OR "relebactam" OR "Avycaz")
Pubmed	((("Drug Resistance, Bacterial"[Mesh]) OR "MDR"[Title/Abstract] OR "Multidrug Resistant"[Title/Abstract]) AND ((("Infections"[Mesh]) OR "bloodstream infection"[Title/Abstract] OR "sepsis"[Title/Abstract] OR "pneumonia"[Title/Abstract] OR "urinary tract infection"[Title/Abstract] OR "catheter-related infection"[Title/Abstract]) AND ((("Beta-Lactamases"[Mesh]) OR "Beta-Lactamase Inhibitor"[Title/Abstract] OR "Beta-Lactam/Beta-Lactamase Inhibitor"[Title/Abstract] OR "ceftazidime-avibactam"[Title/Abstract] OR "avibactam"[Title/Abstract] OR "aztreonam-avibactam"[Title/Abstract] OR "relebactam"[Title/Abstract] OR "tazobactam"[Title/Abstract] OR "imipenem-cilastatin"[Title/Abstract]))
Science Direct	("MDR" OR "Multidrug-Resistant" OR "Drug-Resistant" OR "antibiotic resistance") AND ("infections" OR "bacteremia" OR "bloodstream infections" OR "pneumonia" OR "sepsis" OR "ventilator associated pneumonia" OR "urinary tract infections" OR "catheter associated infections") AND ("Beta-lactamase Inhibitor" OR "ceftazidime-avibactam" OR "avibactam" OR "ceftazidime" OR "aztreonam-avibactam" OR "tazobactam" OR "relebactam" OR "imipenem-cilastatin" OR "beta-lactam inhibitor combination")
ProQuest	("multidrug-resistant" OR "MDR" OR "antibiotic resistant" OR "drug resistant") AND ("infection" OR "bloodstream infection" OR "sepsis" OR "pneumonia" OR "ventilator associated pneumonia" OR "urinary tract infection" OR "catheter related infection") AND ("beta-lactam inhibitor" OR "beta-lactamase inhibitor" OR "ceftazidime-avibactam" OR "aztreonam-avibactam" OR "relebactam" OR "tazobactam" OR "imipenem-cilastatin" OR "avibactam" OR "ceftazidime")
Scopus	(TITLE-ABS-KEY("multidrug resistant" OR "MDR" OR "drug resistant" OR "antibiotic resistant")) AND (TITLE-ABS-KEY("infection" OR "bloodstream infection" OR "sepsis" OR "pneumonia" OR "ventilator-associated pneumonia" OR "urinary tract infection" OR "catheter-related infection")) AND (TITLE-ABS-KEY("beta-lactamase inhibitor" OR "beta-lactam inhibitor" OR "ceftazidime-avibactam" OR "avibactam" OR "ceftazidime" OR "aztreonam-avibactam" OR "relebactam" OR "tazobactam" OR "imipenem-cilastatin"))
BMJ Journal	("Multidrug-resistant bacteria" OR "MDR" OR "Drug Resistant Bacteria") AND ("Infection" OR "Sepsis" OR "Pneumonia" OR "Bloodstream Infection" OR "Urinary Tract Infection" OR "Catheter-Associated Infection") AND ("Beta-lactamase inhibitor" OR "ceftazidime-avibactam" OR "aztreonam-avibactam" OR "tazobactam" OR "relebactam" OR "imipenem-cilastatin" OR "avibactam" OR "ceftazidime")